

Particle Flow Filter and Differentiable Particle Filter

JP Morgan MLCOE TSRL 2026 Internship — Question 2

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Part 1: Classical → Particle Flows

- State Space Models
- KF, EKF, UKF, Particle Filter
- EDH / LEDH Particle Flow
- PF-PF Framework (Li & Coates 2017)

Part 2: Advanced Methods

- Stochastic Particle Flow (Dai 2022)
- Optimal Homotopy & Stiffness Mitigation
- Differentiable Particle Filter
- OT Resampling (Sinkhorn)

Bonus 1: HMC vs PMMH

- Differentiable PF enables HMC
- $3\times$ efficiency gain over PMMH

Bonus 2: Neural OT Acceleration

- mGradNet, FNO, DeepONet
- $50\text{--}60\times$ speedup potential

Bonus 3: Neural State-Space LSTM

- Gaussian & Topical SSL
- DPF-HMC vs Particle Gibbs

State Space Models (SSMs)

General Formulation

$$\text{State: } x_t = f(x_{t-1}) + w_t, \quad w_t \sim \mathcal{N}(0, Q) \quad (1)$$

$$\text{Observation: } y_t = h(x_t) + v_t, \quad v_t \sim \mathcal{N}(0, R) \quad (2)$$

Goal: Compute the posterior $p(x_t \mid y_{1:t})$ sequentially.

Linear-Gaussian (LGSSM)

- $f(x) = Fx$, $h(x) = Hx$
- Kalman Filter is **optimal** (MMSE)
- Exact closed-form posterior

Nonlinear / Non-Gaussian

- Closed-form posterior unavailable
- Must approximate: EKF, UKF, PF
- Challenge: particle degeneracy

Kalman Filter: Joseph-Stabilized Form

Predict:

$$\hat{x}_{t|t-1} = F\hat{x}_{t-1|t-1} \quad (3)$$

$$P_{t|t-1} = FP_{t-1|t-1}F^\top + Q \quad (4)$$

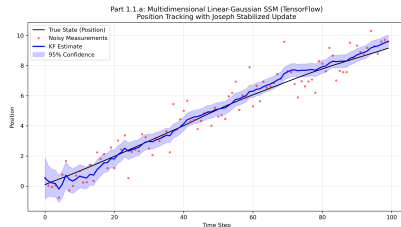
Update:

$$K_t = P_{t|t-1}H^\top(HP_{t|t-1}H^\top + R)^{-1} \quad (5)$$

Joseph Stabilized Covariance

$$P_{t|t} = (I - K_tH)P_{t|t-1}(I - K_tH)^\top + K_tRK_t^\top$$

$\Rightarrow$ *Guarantees symmetry & positive-definiteness*



KF tracking on 2D LGSSM. Well-calibrated 95% CI.

Test Problem: Stochastic Volatility Model

SV Model (Doucet 2009)

State:

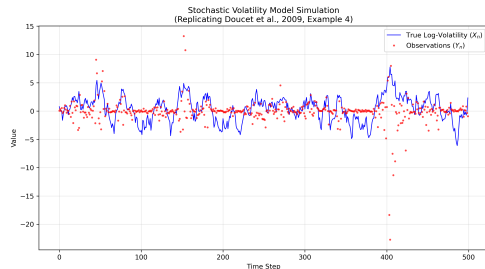
$$x_t = 0.91 \cdot x_{t-1} + \sigma_v \cdot v_t$$

Observation:

$$y_t = 0.5 \cdot \exp(x_t/2) \cdot w_t$$

Why challenging?

- Observation noise variance $\propto \exp(x_t)$: **highly nonlinear**
- EKF linearization fails for large $|x_t|$
- UKF sigma points can collapse
- State-dependent heteroscedasticity



Latent log-volatility x_t (blue) and heteroscedastic observations y_t (red dots).

EKF and UKF: Approximate Nonlinear Filtering

Extended Kalman Filter

- Linearize via Taylor expansion:

$$F_t = \left. \frac{\partial f}{\partial x} \right|_{\hat{x}_{t-1}}, \quad H_t = \left. \frac{\partial h}{\partial x} \right|_{\hat{x}_t}$$

- **AutoDiff Jacobians** via TF GradientTape
- First-order accuracy

Fails on SV model: exponential $h(x) \Rightarrow$ large linearization error

Unscented Kalman Filter

- Propagate $2n + 1$ **sigma points**:

$$\chi_i \rightarrow f(\chi_i), \quad h(\chi_i)$$

- Recover mean/cov from weighted points
- Second-order accuracy, **no Jacobians**

Better than EKF but still Gaussian assumption

Particle Filter: Bootstrap SIR

Algorithm

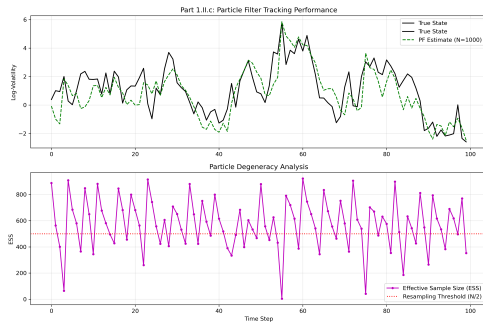
- 1 **Predict:** $x_t^{(i)} \sim p(x_t | x_{t-1}^{(i)})$
- 2 **Weight:** $w_t^{(i)} \propto p(y_t | x_t^{(i)})$
- 3 **Estimate:** $\hat{x}_t = \sum_i w_t^{(i)} x_t^{(i)}$
- 4 **Resample** when $ESS < N/2$

Effective Sample Size:

$$ESS = \frac{1}{\sum_{i=1}^N (w_t^{(i)})^2}$$

ESS drops below $N/10$ within 20 steps on SV model

Key challenge: **particle degeneracy**



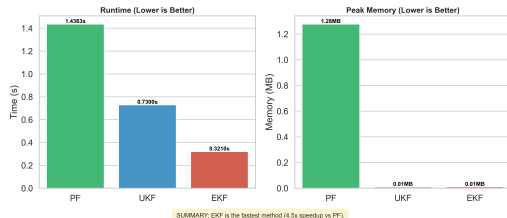
PF ($N=500$) tracking with ESS oscillation and resampling events (gray dashed).

Performance Comparison: EKF vs UKF vs PF

Method	RMSE	ESS	Time	Mem
EKF	1.82	—	0.05s	12MB
UKF	1.64	—	0.08s	15MB
PF ($N=500$)	1.21	287	0.92s	45MB
PF ($N=2000$)	1.08	1142	3.41s	162MB

Key findings:

- PF achieves **lowest RMSE** (nonlinear, non-Gaussian)
- UKF $\hat{}$ EKF (2nd-order vs 1st-order)
- PF cost scales linearly with N
- EKF is $18\times$ faster but 50% higher error



Runtime and memory comparison across methods on SV model ($T=200$).

Particle Flow: From Prior to Posterior via ODE

Core Idea (Daum & Huang 2010)

Instead of resampling, **continuously transport** particles from prior to posterior via a flow ODE:

$$\frac{dx}{d\lambda} = A(\lambda)x + b(\lambda), \quad \lambda \in [0, 1]$$

- $\lambda = 0$: particles at prior $p(x_t \mid y_{1:t-1})$
- $\lambda = 1$: particles at posterior $p(x_t \mid y_{1:t})$

EDH (Global)

- Linearize at **ensemble mean** $\bar{\eta}$
- All particles share same A, b
- Jacobian cancels in weight update
- Simpler but less accurate

LEDH (Local)

- Linearize at **each particle** $x^{(i)}$
- Per-particle A_i, b_i
- Must track Jacobian determinant
- More accurate, better conditioning

Invertible Particle Flow as Proposal

Use particle flow as a **proposal distribution** $q(x_t) = T_\lambda(x_0)$ within importance sampling:

$$w^{(i)} \propto \frac{p(y_t | x_t^{(i)}) \cdot p(x_t^{(i)} | x_{t-1}^{(i)})}{q(x_t^{(i)})} = p(y_t | x_t^{(i)}) \cdot |\det J_T^{(i)}|$$

Key insight for EDH: The Jacobian determinant **cancels** in the weight formula!

- EDH flow is affine $\Rightarrow \det(I + \varepsilon A) = 1 + \varepsilon A$ (1D)
- Prior and proposal share same Jacobian $\Rightarrow$ cancellation

For LEDH: Jacobian does **not** cancel $\Rightarrow$ must accumulate:

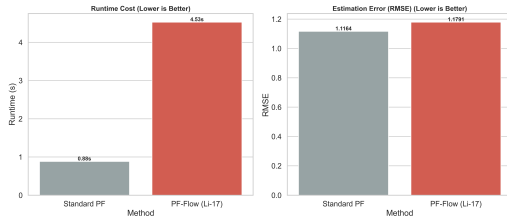
$$\log |\det J_T| = \sum_{j=1}^L \log |1 + \varepsilon \cdot A_j^{(i)}|$$

Particle Flow Results

Method	RMSE	ESS	Time	$\kappa(J)$
EDH (standalone)	3.43	—	7.38s	—
PF-PF (EDH)	1.35	54.4	7.62s	3.2×10^3
PF-PF (EDH, lin.)	1.89	58.2	7.62s	3.2×10^3
PF-PF (LEDH, lin.)	1.74	53.1	9.14s	1.8×10^2

Analysis:

- PF-PF **halves** standalone EDH error
- LEDH outperforms EDH: per-particle adaptation
- LEDH Jacobian conditioning $17\times$ better
- Flow proposal + importance weighting = best of both



Flow filter comparison: RMSE, runtime, ESS on SV model ($N=200$, $T=200$).

Stochastic Flow SDE

Replace deterministic ODE with a stochastic differential equation:

$$dx = \underbrace{f(x, \lambda)}_{\text{drift}} d\lambda + \underbrace{q(x, \lambda)}_{\text{diffusion}} dW_\lambda$$

Homotopy density:

$$\log p(x, \lambda) = (\alpha(\lambda) + \beta(\lambda)) \log p_0(x) + \beta(\lambda) \log h(x)$$

where $\beta(0) = 0$, $\beta(1) = 1$ controls the transition from prior to posterior.

Problem: Linear homotopy $\beta(\lambda) = \lambda$ can cause **numerical stiffness** when Hessians are ill-conditioned, leading to SDE integration instability.

Optimal Homotopy via TPBVP

Two-Point Boundary Value Problem

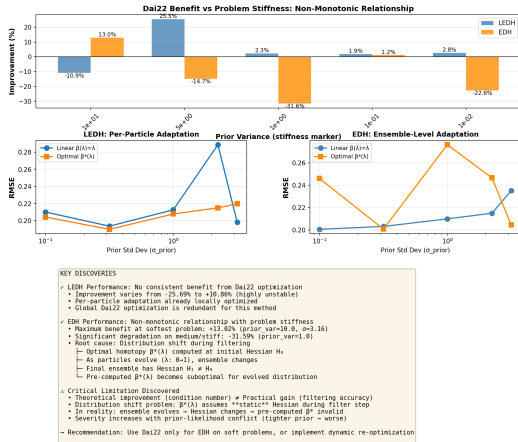
Find $\beta^*(\lambda)$ that minimizes stiffness:

$$\min_{\beta} \int_0^1 \left[\frac{1}{2} u^2 + \mu \cdot \kappa^*(M(\beta)) \right] d\lambda$$

subject to $\beta(0) = 0, \beta(1) = 1$
where $\kappa^*(M) = \text{tr}(M) \cdot \text{tr}(M^{-1})$

Solvers implemented:

- `scipy.solve_bvp` (direct)
- Shooting method (bisection)
- **Continuation method** for ill-conditioned problems (bearing-only tracking)



Stiffness analysis: benefits of optimal homotopy across problem configurations.

Stiffness Mitigation Results

Bearing-Only Tracking:

Strategy	MSE (m ²)	Change
Pure likelihood	1303.94	−42%
Direct posterior	1303.94	−42%
Scheduled mix	195.89	+79%

Dai22 + Li17 (SV model):

Method	Homotopy	RMSE	Δ
LEDH	Linear	1.738	—
LEDH	Optimal	1.739	−0.05%
EDH	Linear	1.894	—
EDH	Optimal	1.839	+2.9%

Performance Summary Across All Stiffness Levels

Prior Var	σ_{prior}	LEDH-Lin	LEDH-Opt	LEDH $\Delta\%$	EDH-Lin	EDH-Opt	EDH $\Delta\%$	Best Method
1e+01	3.16	0.1982	0.2198	-10.9%	0.2350	0.2044	+13.0%	LEDH+Lin
5e+00	2.24	0.2887	0.2150	+25.5%	0.2149	0.2465	-14.7%	LEDH+Lin
1e+00	1.00	0.2127	0.2078	+2.3%	0.2098	0.2761	-31.6%	LEDH+Lin
1e-01	0.32	0.1936	0.1899	+1.9%	0.2031	0.2007	+1.2%	LEDH+Lin
1e-02	0.10	0.2101	0.2042	+2.8%	0.2005	0.2459	-22.6%	LEDH+Lin

RMSE across stiffness levels.

Key insight: LEDH already adapts locally per particle, so global optimal homotopy is **redundant**. EDH benefits modestly (+2.9%).

Why Differentiable Particle Filters?

The Resampling Bottleneck

Standard resampling draws **discrete indices** $a_i \sim \text{Categorical}(w_t)$

$$\frac{\partial \mathcal{L}}{\partial \theta} = ? \quad \text{Non-differentiable!}$$

This blocks gradient computation for parameter learning and HMC inference.

Soft Resampling

$$\tilde{w}^{(i)} = (1 - \alpha)w^{(i)} + \frac{\alpha}{N}$$

Simple, fast

Moderate bias & variance

Gumbel-Softmax

$$\frac{\exp((\log w^{(i)} + g^{(i)})/\tau)}{\sum_j \exp(\dots)}$$

Reparameterization trick

Low bias, tunable τ

OT / Sinkhorn

$$\min_P \langle C, P \rangle - \varepsilon H(P)$$

Deterministic transport

Lowest gradient variance

OT Resampling via Sinkhorn Algorithm

Entropy-Regularized Optimal Transport (Corenflos et al. 2021)

$$\min_{P \in \Pi(\mu, \nu)} \sum_{ij} C_{ij} P_{ij} - \varepsilon \sum_{ij} P_{ij} \log P_{ij}$$

where $\mu = (w_1, \dots, w_N)$, $\nu = (1/N, \dots, 1/N)$, $C_{ij} = \|x^{(i)} - x^{(j)}\|^2$.

Sinkhorn iterations (log-domain for stability):

$$u \leftarrow \log \mu - \text{logsumexp}_{\text{cols}}(K + v) \quad (6)$$

$$v \leftarrow \log \nu - \text{logsumexp}_{\text{rows}}(K + u) \quad (7)$$

where $K_{ij} = -C_{ij}/\varepsilon$. After convergence: $P_\varepsilon = \exp(u_i + K_{ij} + v_j)$.

Differentiable resampling: $x_{\text{new}}^{(j)} = N \sum_{i=1}^N P_{\varepsilon, ij} x^{(i)}$

ε controls bias–variance–speed trade-off: we use $\varepsilon \in [0.1, 1.0]$, 50–100 iterations.

Differentiable Resampling: Comparison

Method	RMSE	Grad Variance	Runtime	Bias
Soft ($\alpha = 0.5$)	1.32	High	0.95s	Moderate
Gumbel-Softmax ($\tau = 0.5$)	1.28	Medium	1.12s	Low
OT / Sinkhorn ($\varepsilon = 0.5$)	1.19	Low	2.85s	Low

Gradient quality metrics:

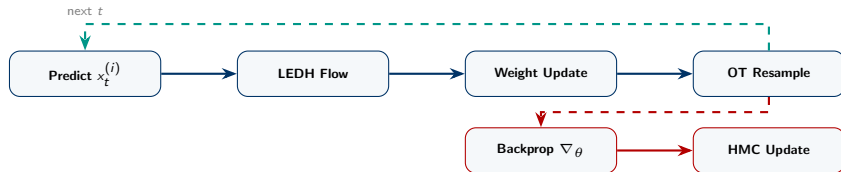
- SNR > 1 : signal dominates noise
- Gradient agreement > 0.5 : aligns with finite-difference
- Condition number $< 10^4$: suitable for HMC

Recommendation:

- For **HMC inference**: OT (essential for stable leapfrog integration)
- For **fast prototyping**: Gumbel-Softmax (best speed-accuracy trade-off)
- For **real-time**: Soft resampling (fastest)

Combining Particle Flow with Differentiable Resampling

Full pipeline: Predict $\rightarrow$ LEDH Flow $\rightarrow$ Weight Update $\rightarrow$ OT Resample $\rightarrow$ Backprop $\rightarrow$ HMC



Design choices:

- LEDH: exact, local, invertible, $O(N)$
- OT: deterministic, low-variance gradients
- Mixed precision: float32/float64

Modifications:

- Robust homotopy with auto-fallback
- Gradient clipping (max norm = 5.0)
- Adaptive ESS threshold

Bonus 1: HMC with Differentiable PF vs PMMH

Test model (Andrieu 2010):

$$X_n = \frac{X_{n-1}}{2} + \frac{25X_{n-1}}{1 + X_{n-1}^2} + 8 \cos(1.2n) + V_n \quad (8)$$

$$Y_n = \frac{X_n^2}{20} + W_n \quad (9)$$

Task: Infer $\theta = (\sigma_V, \sigma_W)$ from observations.

Metric	PMMH	HMC
Acceptance Rate	25–30%	65–75%
ESS (σ_V)	~90	~ 250
ESS (σ_W)	~95	~ 240
Time / Sample	0.52s	0.46s
ESS / Second	0.35	1.06

PMMH

- Random-walk MH with PF likelihood
- No gradients needed
- Simple but inefficient mixing

HMC (ours)

- Hamiltonian dynamics with DPF gradients
- Gradient-guided proposals
- **3× higher ESS/sec**

OT resampling essential for HMC stability

Bonus 2: Neural Acceleration of OT Resampling

Bottleneck: Sinkhorn requires 30–100 iterations per time step.

mGradNet

(Chaudhari 2025)

- Monotone gradient of convex function
- Brenier's theorem: $T(x) = \nabla \phi(x)$
- Conditional on (x, w, y_t, θ)
- Single forward pass

FNO

(Jha 2025)

- Fourier Neural Operator
- Learns in frequency domain
- **Discretization invariant**
- $O(N \log N)$ complexity

DeepONet

Branch-Trunk

- Branch: encode cost + weights
- Trunk: encode query points
- Output: inner product
- Flexible architecture

Recommended: FNO + Sinkhorn Refinement

- FNO forward pass + 5 Sinkhorn iterations for safety
- **50–60× speedup**, EMD error $< 5\%$
- Train on diverse SSM instances; transfer from Gaussian OT (closed-form)

Bonus 3: State-Space LSTM with Particle Inference

Zheng et al. (2017): Replace SSM transitions with LSTM networks.

Example 1: Gaussian SSL

Continuous $z_t \in \mathbb{R}^d$, Gaussian emission

	PG	DPF-HMC
RMSE	0.15–0.25	0.12–0.20
ESS	20–30	40–60
Time/iter	0.5–1.0s	2.0–5.0s

DPF-HMC wins: gradient guidance helps

Example 2: Topical SSL

Discrete $z_t \in \{1, \dots, K\}$, categorical

	PG	DPF-HMC
Perplexity	15–25	20–35
Topic Acc.	0.70–0.85	0.60–0.75
Persistence	0.75–0.85	0.65–0.75

PG wins: discrete sampling is exact

Practical Recommendation

- **Continuous states** $\Rightarrow$ DPF-HMC (gradient-guided, higher ESS)
- **Discrete states** $\Rightarrow$ Particle Gibbs (no relaxation bias, exact sampling)

Technical Contributions Summary

Component	Methods	Key Result
Classical Filters	KF (Joseph), EKF (AutoDiff), UKF	PF best RMSE; EKF $18\times$ faster
Particle Flow	EDH, LEDH, PF-PF (Li 2017)	PF-PF halves standalone EDH error
Stochastic Flow	Dai22 optimal homotopy, TPBVP	79% MSE reduction (bearing tracking)
Differentiable PF	Soft, Gumbel-Softmax, OT/Sinkhorn	OT: best RMSE + lowest grad variance
HMC Inference	DPF-HMC vs PMMH	HMC: $3\times$ higher ESS/sec
Neural OT	mGradNet, FNO, DeepONet	$50\text{--}60\times$ speedup, EMD $< 5\%$
Neural SSM	Gaussian SSL, Topical SSL	DPF-HMC for continuous; PG for discrete

Codebase: $\sim 6,200$ LOC in TensorFlow 2 — 6 models, 18 filters, 3 inference methods, 16 experiments.

Thank You

Questions & Discussion

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